

REPUBLIC OF SOUTH AFRICA

PATENTS ACT, 1978

APPLICATION FOR A PATENT AND ACKNOWLEDGEMENT OF RECEIPT

[Section 30 (1)-Regulation 22]

The granting of a patent is hereby requested by the undermentioned applicant on the basis of the present application.

Official Application No.			Applicant's or Agent's Reference	
21	01	2026/05473	6895LPS	

71	Full Name(s) of Applicant(s)			
VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India				

54	Title of invention			
AN INTELLIGENT FRUIT AND VEGETABLE IDENTIFICATION AND QUALITY CLASSIFICATION SYSTEM				

The applicant claims priority as set out on the accompanying Form P.2. The earliest priority claimed is				
COUNTRY:		NUMBER:		DATE:

This application is for a patent of addition to patent application No.				
21	01			

This application is a fresh application in terms of section 37 and based on Application No.				
21	01			

This application is accompanied by:				
X	1.	A single copy of a complete specification of 16 pages.		
X	2.	Drawings of 2 sheet(s).		
X	3.	Publication particulars and abstract(Form P8)		
X	4.	A copy of a figure of the drawing (if any) for the abstract		
X	5.	Assignment of invention		
	6.	Certified priority document(s)		
	7.	Translation(s) of the priority document(s)		
	8.	Assignment of priority rights		
	9.	A copy of the Form P.2 and the specification of S.A Patent Application (if applicable).		
X	10.	A declaration and power of attorney on Form P3		
X	11.	Statement on the use of indigenous Biological Resource, Genetic Resource, Traditional Knowledge or Use on Form P26		

74	Address of Service:			
Wolmarans and Susan Inc. 337 Surrey Avenue, Randburg, Gauteng, 2194 SOUTH AFRICA				

Dated this 21st day of May 2026

Digitally signed by : Leon Pierre Susan

.....
Signature of Applicant(s)

This is returned to the applicant's
address for service as proof of lodging.

RECEIVED
Official Date Stamp
..... Registrar of Patents

REPUBLIC OF SOUTH AFRICA

REGISTER OF PATENTS

PATENTS ACT, 1978

Official application No.		Lodging date: Provisional		Acceptance date	
21	01	2026/05473		22	
International classification		Lodging date: Complete		Granted date	
51	G06N	23	2026/05/21		
71 Full name(s) of applicant(s)/Patentee(s):					
VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India					
71 Applicant substituted:				Date registered	
71 Assignee(s):				Date registered	
72 Full name(s) of inventor(s):					
BELHE, Sayali Kulkarni MUNNALURI, Sri Sai Preethi VYAS, Anvesha BAM, Shlok					
Priority claimed:		Country	Number	Date	
54 Title of invention					
AN INTELLIGENT FRUIT AND VEGETABLE IDENTIFICATION AND QUALITY CLASSIFICATION SYSTEM					
Address of applicant(s)/patentee(s):					
666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037 INDIA					
74 Address for service					
Wolmarans and Susan Inc. 337 Surrey Avenue, Randburg, Gauteng, 2194 SOUTH AFRICA Reference No. 6895LPS					
61 Patent of addition No.			Date of any change		
Fresh application based on.			Date of any change		

REPUBLIC OF SOUTH AFRICA
 PATENTS ACT, 1978
COMPLETE SPECIFICATION
[Section 30(1) – Regulation 28]

OFFICIAL APPLICATION NO.

21	01	2026/05473
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LODGING DATE

22	2026/05/21
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INTERNATIONAL CLASSIFICATION

51	G06N
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FULL NAME(S) OF APPLICANT(S)

71	VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India
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FULL NAME(S) OF INVENTORS(S)

72	1. BELHE, Sayali Kulkarni 2. MUNNALURI, Sri Sai Preethi 3. VYAS, Anvesha 4. BAM, Shlok
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TITLE OF INVENTION

54	AN INTELLIGENT FRUIT AND VEGETABLE IDENTIFICATION AND QUALITY CLASSIFICATION SYSTEM
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REPUBLIC OF SOUTH AFRICA
PATENTS ACT, 1978
PUBLICATION PARTICULARS AND ABSTRACT
[Section 32(3)(a) – Regulation 2291(g) AND 31]

OFFICIAL APPLICATION NO.		LODGING DATE	ACCEPTANCE DATE
21	01	22	47
2026/05473		2026/05/21	

INTERNATIONAL CLASSIFICATION	NOT FOR PUBLICATION
51 G06N	CLASSIFIED BY: Wolmarans and Susan Inc.

FULL NAME(S) OF APPLICANT(S)	
71	VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India

FULL NAME(S) OF INVENTORS(S)	
72	1. BELHE, Sayali Kulkarni 2. MUNNALURI, Sri Sai Preethi 3. VYAS, Anvesha 4. BAM, Shlok

COUNTRY	NUMBER	DATE
33	31	32

TITLE OF INVENTION	
54	AN INTELLIGENT FRUIT AND VEGETABLE IDENTIFICATION AND QUALITY CLASSIFICATION SYSTEM

57	The present invention relates to an intelligent fruit and vegetable detection and quality classification system configured to identify, analyze, and grade agricultural produce using digital image processing and machine learning techniques. The system comprises an image acquisition unit for capturing produce images, a preprocessing module for noise removal and background normalization, a segmentation module for object isolation, and a classification engine based on trained machine learning or deep learning models. The system identifies fruit and vegetable types, evaluates freshness, detects surface defects, and assigns quality grades in real time. The system operates with high accuracy and low computational complexity and supports both online and offline processing modes. The invention provides a cost-effective, scalable, and automated solution for quality control in agriculture, food processing, and retail industries.
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